Capillary leak syndrome (CLS)

Capillary leak syndrome has been reported in receiving gemcitabine as single agent or in combination with chemotherapeutic agents. The condition is usually treatable if recognised early and managed appropriately, but fatal cases have been reported. The condition involves systemic capillary hyperpermeability during which fluid and proteins from the intravascular space leak into the interstitium. The clinical features include generalised oedema, weight gain, hypoalbuminaemia, severe hypotension, acute renal impairment and pulmonary oedema. Gemcitabine should be discontinued and supportive measures implemented if capillary leak syndrome develops during therapy. Capillary leak syndrome can occur in later cycles and has been associated in the literature with respiratory distress syndrome.

Posterior reversible encephalopathy syndrome (PRES)

Reports of posterior reversible encephalopathy syndrome (PRES) with potentially severe consequences have been reported in receiving gemcitabine as single agent or in combination with other chemotherapeutic agents. Acute hypertension and seizure activity were reported in most gemcitabine experiencing PRES, but other symptoms such as headache. Lethargy, confusion and blindness could also be present. Diagnosis is optimally confirmed by magnetic resonance imaging (MRI). PRES was typically reversible with appropriate supportive measures. Gemcitabine should be permanently discontinued and supportive measures implemented, including blood pressure control and anti-seizure therapy, if PRES develops during therapy.

Pulmonary

Pulmonary effects, sometimes severe (such as pulmonary oedema, interstitial pneumonitis or adult respiratory distress syndrome (ARDS)) have been reported in association with gemcitabine therapy. The aetiology of these effects is unknown. If such effects develop, consideration should be made to discontinuing gemcitabine therapy. Early use of supportive care measure may help ameliorate the condition.

Renal

Haemolytic uraemic syndrome

Clinical findings consistent with the haemolytic uraemic syndrome (HUS) were rarely reported in receiving gemcitabine Gemcitabine should be discontinued at the first signs of any evidence of microangiopathic haemolytic anaemia, such as rapidly falling haemoglobin with concomitant thrombocytopaenia, elevation of serum bilirubin, serum creatinine, blood urea nitrogen, or LDH. Renal failure may not be reversible with discontinuation of therapy and dialysis may be required.

Fertility

In fertility studies gemcitabine caused hypospermatogenesis in male mice .Therefore, men being treated with gemcitabine are advised not to father a child during and up to 6 months after treatment and to seek further advice regarding cryoconservation of sperm prior to treatment because of the possibility of infertility due to therapy with gemcitabine .

Sodium

Gemcitabine 200 mg powder for solution for infusion contains 3.5 mg (<1 mmol) sodium per vial. This should be taken into consideration on a controlled sodium diet.

Gemcitabine 1 g powder for solution for infusion contains 17.5 mg (< 1 mmol) sodium per vial. This should be taken into consideration by patients on a controlled sodium diet.

m) Interactions with other medicaments

No specific interaction studies have been performed .

Radiotherapy

Concurrent (given together or ≤☐7 days apart) - Toxicity associated with this multimodality therapy is dependent on many different factors, including dose of gemcitabine, frequency of gemcitabine administration, dose of radiation, radiotherapy planning technique, the target tissue, and target volume. radiosensitising activity. In a single trial, where gemcitabine at a dose of 1,000 mg/m^2 was administered concurrently for up to 6 consecutive weeks with therapeutic thoracic radiation to patients with non-small cell lung cancer, significant toxicity in the form of severe, and potentially life threatening mucositis, especially oesophagitis, and pneumonitis was observed, particularly in patients receiving large volumes of radiotherapy [median treatment volumes 4,795 cm^3]. Studies done subsequently have suggested that it is feasible to administer gemcitabine at lower doses with concurrent radiotherapy with predictable toxicity, such as a phase II study in non-small cell lung cancer, where thoracic radiation doses of 66 Gy were applied concomitantly with an administration with gemcitabine (600 mg/m^2, four times) and cisplatin (80 mg/m2 twice) during 6 weeks. The optimum regimen for safe administration of gemcitabine with therapeutic doses of radiation has not yet been determined in all tumour types.

Non-concurrent (given >7 days apart)- Analysis of the data does not indicate any enhanced toxicity when gemcitabine is administered more than 7 days before or after radiation, other than radiation recall. Data suggest that gemcitabine can be started after the acute effects of radiation have resolved or at least one week after radiation.

Radiation injury has been reported on targeted tissues (e.g. oesophagitis, colitis, and pneumonitis) in association with both concurrent and non-concurrent use of gemcitabine.

Others

Yellow fever and other live attenuated vaccines are not recommended due to the risk of systemic, possibly fatal, disease, particularly in immunosuppressed patients.

n) Side Effects

Very common (more than 10% of people develop them) adverse effects include difficulty breathing, low white and red blood cells counts and low platelet counts, vomiting and nausea, elevated transaminases, rashes and itchy skin, hair loss, blood and protein in urine, flu-like symptoms, and edema.

Common adverse effects (occurring in 1–10% of people) include fever, loss of appetite, headache, difficulty sleeping, tiredness, cough, runny nose, diarrhea, mouth and lip sores, sweating, back pain, and muscle pain.

The following table of undesirable effects and frequencies

System Organ Class	Incident	Frequency grouping
Blood and lymphatic system disorders	Very common	Leucopaenia (Neutropaenia Grade 3 = 19.3 %; Grade 4 = 6 %). Bone-marrow suppression is usually mild to moderate and mostly affects the granulocyte count • Thrombocytopaenia • Anaemia
	Common	• Febrile neutropaenia
	Very rare	• Thrombocytosis • Thrombotic microangiopathy
Infections and infestations	Very Rare	• Anaphylactoid reaction
Metabolism and nutrition disorders	Common	• Anorexia
Nervous system disorders	Common	• Headache • Insomnia • Somnolence
	Very rare	• Posterior reversible encephalopathy syndrome
Cardiac disorders	Rare	• Myocardial infarct
Vascular disorders	Rare	• Hypotension
	Very rare	• Capillary leak syndrome
Respiratory, thoracic and mediastinal disorders	Very common	• Dyspnoea –usually mild and passes rapidly without treatment
	Common	• Cough • Rhinitis
	Uncommon	• Interstitial pneumonitis • Bronchospasm –usually mild and transient but may require parenteral treatment
Gastrointestinal disorders	Very common	• Vomiting • Nausea
	Common	• Diarrhoea • Stomatitis and ulceration of the mouth • Constipation
Hepatobiliary disorders	Very common	• Elevation of liver transaminases (AST and ALT) and alkaline phosphatase
	Common	• Increased bilirubin
	Rare	• Increased gamma-glutamyl transferase (GGT)
Skin and subcutaneous tissue disorders	Very common	• Allergic skin rash frequently associated with pruritus • Alopecia
	Common	• Itching • Sweating
	Rare	• Ulceration • Vesicle and sore formation • Scaling
	Very rare	• Severe skin reactions, including desquamation and bullous skin eruptions
	Not known	• Pseudocellulitis
Musculoskeletal and connective tissue disorders	Common	• Back pain • Myalgia
Renal and urinary disorders	Very common	• Haematuria • Mild proteinuria
General disorders and administration site conditions	Very common	• Influenza-like symptoms - the most common symptoms are fever, headache, chills, myalgia, asthenia and anorexia. Cough, rhinitis, malaise, perspiration and sleeping difficulties have also been reported. • Oedema/peripheral oedema including facial oedema. Oedema is usually reversible after stopping treatment
	Common	• Fever • Asthenia • Chills
	Rare	• Injection site reactions-mainly mild in nature

o) Symptoms and Treatment of Overdose

There is no known antidote for overdose of Gemcitabine hydrochloride. Doses as high as 5700 mg/m 2 have been administered by intravenous infusion over 30 minutes every 2 weeks with clinically acceptable toxicity. In the event of suspected overdose, the patient should be monitored with appropriate blood counts and receive supportive therapy, as necessary.

p) Effects on Ability to drive and use Machine

The amount of alcohol in this medicinal product may impair the ability to drive or use machines.

No studies on the effects on the ability to drive and use machines have been performed. However, Gemcitabine has been reported to cause mild to moderate somnolence, especially in combination with alcohol consumption. Patients should be cautioned against driving or operating machinery until it is established that they do not become somnolent.

q) Preclinical Safety Data

Not Applicable for Generic Product.

r) Instructions for use Handling:

Handling:

The normal safety precautions for cytostatic agents must be observed when preparing and disposing of the infusion solution. Handling of the solution for infusion should be done in a safety box and protective coats and gloves should be used. If no safety box is available, the equipment should be supplemented with a mask and protective glasses.

If the preparation comes into contact with the eyes, this may cause serious irritation.

The eyes should be rinsed immediately and thoroughly with water. If there is lasting irritation, a doctor should be consulted. If the solution is spilled on the skin, rinse thoroughly with water.

Instructions for reconstitution (and further dilution, if performed):

The only approved diluent for reconstitution of Gemcitabine sterile powder is sodium chloride 9 mg/mL (0.9%) solution for injection (without preservative). Due to solubility considerations, the maximum concentration for Gemcitabine upon reconstitution is 40 mg/mL. Reconstitution at concentrations greater than 40 mg/mL may result in incomplete dissolution and should be avoided.

1. Use aseptic technique during the reconstitution and any further dilution of Gemcitabine for intravenous infusion administration.

2. To reconstitute, add 5 mL of sterile sodium chloride 9 mg/mL (0.9 %) solution for injection, without preservative, to the 200 mg vial or 25 mL sterile sodium chloride 9 mg/mL (0.9 %) solution for injection, without preservative, to the 1 g vial. The total volume after reconstitution is 5.26 mL (200 mg vial) or 26.3 mL (1 g vial) respectively. This yields a Gemcitabine concentration of 38 mg/mL, which includes accounting for the displacement volume of the lyophilised powder. Shake to dissolve.

Further dilution with sterile sodium chloride 9 mg/mL (0.9 %) solution for injection, without preservative can be done. Reconstituted solution is a clear colourless to light straw-coloured solution.

3. Parenteral medicinal products should be inspected visually for particulate matter and discolouration prior to administration. If particulate matter is observed, do not administer.

Any unused product or waste material should be disposed of in accordance with local requirements.

s) Special precautions for storage

Unopened vial: Store below 30°C

After reconstitution and After Dilution: The reconstituted solution is stable for 6 hours at a controlled room temperature 30+/- 2 °C/75+/- 5%. Do not refrigerate as crystallization can occur.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

t) Shelf life

3 years

After reconstitution and After Dilution : The reconstituted solution is stable for 6 hours at a controlled room temperature 30+/- 2 °C/75+/- 5%. Do not refrigerate as crystallization can occur.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

u) Dosage forms and packaging available

The Gemcitabine Powder for solution for Infusion is primarily packed in 10mL tubular clear(Type –I glass) with 20mm collar flat bottom vial with 20mm Grey bromobutyl (single slotted)lyo rubber stopper and sealed with 20mm easy to open C/L Alu. Seal with matt finish plastic german blue color button.

The Gemcitabine Powder for solution for Infusion is primarily packed in 50mL tubular clear(Type –I glass) with 20mm collar flat bottom vial with 20mm Grey bromobutyl (single slotted)lyo rubber stopper and sealed with 20mm easy to open C/L Alu. Seal with matt finish plastic german blue color button.

v) Therapeutic code

L01BC05

w) Name and address of manufacturer/ product registration holder

Manufactured By:

SP Accure Labs Pvt. Ltd.

Plot No.12, Biotech Park, Phase 2,
Lalgadi Malakpet, Shameerpet (M),
Medchal Dist, Telangana-500 101,
INDIA.

x) Product registration holder

Healol Pharmaceuticals Sdn.Bhd.
74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju,
53300 Kuala Lumpur, Malaysia

y) Date of revision of PI

23, January, 2023

TS/DRUGS/3/2016

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